Predicting Market Correlation Structure as a Spatiotemporal Video Forecasting Problem

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Abstract. We propose a novel formulation of stock market correlation dynamics as a spatiotemporal video forecasting problem. Rather than predicting pairwise correlations in isolation or using correlation as a static graph feature, we treat the full $N \times N$ cross-asset correlation matrix as a sequence of grayscale image frames and apply deep learning architectures originally developed for video understanding. We introduce three key methodological contributions: (1) Fiedler vector reordering of the correlation matrix, which places correlated assets spatially adjacent and makes cluster structure detectable by convolutional kernels; (2) adaptive temporal resolution training, inspired by video frame interpolation techniques, which generates sub-daily correlation snapshots for volatile market periods to provide richer signal during regime transitions; and (3) a directional learning rate scheduler with weight rewinding, which adapts the learning rate based on the direction of progress toward the prediction target rather than elapsed time. Experiments on 95 Nifty 100 stocks using 5-minute bar data spanning July 2021 to April 2026 demonstrate that the proposed approach achieves 44.4% MSE reduction over the persistence baseline, 73.0% directional accuracy on 5-day-ahead correlation change prediction, and a Pearson correlation of 0.294 between predicted and actual future correlation matrices. These results establish that market correlation structure contains learnable spatiotemporal patterns and that video-based deep learning is a viable and novel framework for correlation matrix forecasting.

Keywords: Stock market correlation · Spatiotemporal forecasting · Convolutional neural networks · Correlation matrix prediction · Transfer learning · Financial deep learning · Fiedler vector · Adaptive resolution

1 Introduction

The pairwise correlation structure of financial assets is a fundamental quantity in risk management, portfolio construction, and systemic risk analysis. Yet while a large body of work exists on predicting individual asset returns, relatively little attention has been paid to predicting the *future structure* of the correlation matrix itself — that is, forecasting how the entire pattern of co-movement relationships between assets will evolve over time.

Existing approaches to correlation prediction fall broadly into two camps. The first addresses pairwise correlation prediction in isolation: given the historical correlation between two specific assets, predict whether it will strengthen or weaken [1,2]. These approaches treat each pair independently and cannot capture market-wide structural shifts that simultaneously reshape many correlations. The second camp uses the correlation matrix as a static graph for downstream tasks such as return prediction or portfolio optimization [5,6], without attempting to forecast how the graph will evolve.

The gap we address is: *given the current market correlation structure, what will the full correlation matrix look like in H trading days?* This question has significant practical implications. A model that predicts future correlation structure can simultaneously forecast which asset clusters will form or dissolve, where contagion will spread in a market shock, and how portfolio diversification will change — all from a single prediction.

Our key insight is that this problem maps naturally to spatiotemporal video forecasting. At each trading day, the $N \times N$ correlation matrix constitutes a single frame, where each pixel (i, j) represents the current correlation between assets i and j. A sequence of L frames forms a video clip capturing the temporal evolution of market structure. Brightening pixels indicate strengthening correlations; fading pixels indicate weakening ones. Cluster formation appears as the emergence of bright diagonal blocks; cluster dissolution as these blocks fading.

This framing enables direct application of convolutional neural network (CNN) architectures to detect spatial cluster structure within each frame, and recurrent or three-dimensional convolutional architectures to capture how that structure evolves across frames — precisely the inductive biases needed for our prediction task.

The main contributions of this paper are:

1. A novel formulation of correlation matrix forecasting as spatiotemporal video prediction, enabling direct application of video deep learning architectures to financial markets.
2. Fiedler vector reordering of the correlation matrix, which transforms cluster structure into spatially coherent image patterns detectable by convolutional kernels.
3. Adaptive temporal resolution training, which generates sub-daily correlation snapshots for volatile periods, providing richer training signal during market regime transitions.
4. A directional learning rate scheduler with weight rewinding, which actively rewards progress toward the prediction target and recovers from backward steps.
5. Empirical demonstration that future market correlation structure is predictable, with 44.4% MSE reduction over persistence and 73.0% directional accuracy on 5-day-ahead forecasts.

2 Related Work

2.1 Pairwise Correlation Prediction

Dynamic Conditional Correlation (DCC) models [1] and their extensions represent the classical approach to time-varying correlation estimation. Deep learning approaches have attempted to improve on these by capturing non-linear dynamics [4], but still operate at the pair level and cannot capture market-wide structural transitions.

2.2 Graph-Based Stock Market Models

A growing literature uses the correlation matrix to construct a stock graph and applies graph neural networks for return prediction [5,6,7]. These approaches use correlation as a fixed graph topology rather than as a prediction target. Graph WaveNet [8] and related spatiotemporal GNN approaches predict signals on a fixed graph; our work instead predicts the graph itself.

2.3 Covariance Matrix Forecasting

Classical approaches use shrinkage estimators or factor models [2,3] and focus on estimation accuracy rather than dynamic forecasting. Neural network approaches to covariance forecasting [9] have appeared recently but operate on low-dimensional factor representations rather than the full asset-level structure.

2.4 Spatiotemporal Deep Learning

Video understanding architectures including 3D CNNs [10], ConvLSTM [11], and transfer learning from large-scale image datasets [12] have demonstrated strong performance on tasks requiring joint spatial and temporal reasoning. To our knowledge, these architectures have not been applied to financial correlation matrix forecasting.

2.5 Spectral Graph Theory in Finance

The Fiedler vector (second Laplacian eigenvector) provides the optimal bisection of a graph [13] and has been used for hierarchical clustering of financial assets [14,15]. Our work is the first to use it for spatial reordering of the correlation matrix to enable convolutional processing.

3 Prior Approach: Pair-Level Prediction and Its Ceiling

Before developing the correlation matrix video formulation, we investigated pair-level relationship prediction as a natural starting point. Given the historical correlation time series between two assets, can a model predict whether their relationship will strengthen, remain stable, or weaken over the next trading day?

3.1 Experimental Setup

Using the same 95 Nifty 100 stocks and 5-minute bar data, we computed three variants of the correlation score R(t) for all 4,465 unique asset pairs: a single-day rolling window (75 bars), a weekly rolling window (375 bars), and an exponentially weighted moving average ($\lambda = 0.94$). Velocity and acceleration of each score were derived via finite differences, yielding 11 features per pair per timestep. India VIX and Nifty 50 return were appended as market context, giving 12 features total.

Labels were defined as a 3-class problem (WEAKEN / STABLE / STRENGTHEN) based on the change in EWMA correlation over a 75-bar horizon, thresholded at the 33rd and 67th training percentiles.

3.2 Results and Ceiling

Table 1 reports validation accuracy for all pair-level models. All architectures converge to approximately 56% accuracy regardless of complexity.

Table 1. Pair-level relationship prediction (3-class). All models converge to the same ceiling regardless of architecture complexity.

Model	Val Accuracy	vs. Random
Random baseline	33.3%	—
XGBoost (12 features)	55.6%	+22.3pp
LSTM (12 features)	55.8%	+22.5pp
Ensemble (XGB+LSTM)	56.1%	+22.8pp
Transformer + GNN	55.5%	+22.2pp

3.3 Why the Ceiling Exists

The convergence of all architectures to ≈56% reveals a fundamental limitation of the pair-level formulation. When an external shock reshapes the market, it simultaneously affects all pair correlations. A model observing only one pair's history cannot distinguish between an idiosyncratic relationship change and a market-wide structural shift. Adding VIX as a market context feature provided no improvement, confirming that market context needs to be modeled as a spatial structure — the full correlation matrix — not as a scalar feature. This insight directly motivated the reformulation described in this paper.

4 Problem Formulation

4.1 Correlation Matrix Video

Let $\mathbf{r}_t \in \mathbb{R}^N$ denote the vector of log returns for N assets at time t. We compute the EWMA correlation matrix at each timestep with decay factor $\lambda = 0.94$,

sampled at daily resolution (every 75 five-minute bars), producing a sequence of $N \times N$ correlation matrices $\{R_1, R_2, \ldots, R_T\}$ with $R_t^{(ij)} \in [-1, +1]$.

4.2 Fiedler Vector Reordering

The raw correlation matrix has no inherent spatial structure. We reorder assets using the Fiedler vector [13] — the eigenvector corresponding to the second-smallest eigenvalue of the graph Laplacian $L = D - A$, where $A_{ij} = \max(0, \bar{R}^{(ij)})$ is the mean positive correlation and D is the degree matrix. Sorting assets by Fiedler vector values places strongly correlated assets spatially adjacent, producing coherent diagonal block structure as shown in Fig. 1. The reordering is computed once and applied consistently to all frames, ensuring stable spatial positions across time.

4.3 Prediction Task

Given a lookback sequence of L reordered frames $\mathbf{X}_t = \{R_{t-L+1}, \ldots, R_t\}$, the model predicts the future correlation matrix:

$$\hat{R}_{t+H} = f_\theta(\mathbf{X}_t) \tag{1}$$

where H is the prediction horizon. We use $L = 20$ (20 trading days) and $H = 5$ (5 trading days ahead). The training objective is weighted mean squared error:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} w_t \cdot \frac{1}{N^2} \|\hat{R}_{t+H} - R_{t+H}\|_F^2 \tag{2}$$

where w_t is a volatility-based sample weight and $\| \cdot \|_F$ denotes the Frobenius norm.

5 Methodology

5.1 Fiedler Vector Reordering: From Arbitrary Matrix to Structured Image

The raw correlation matrix R_t has no inherent spatial structure — the ordering of assets along rows and columns is arbitrary, determined by listing order rather than financial relationships. Under random ordering, correlated assets are scattered across the matrix and cluster patterns appear as dispersed bright patches rather than contiguous blocks. Convolutional kernels, which detect local spatial patterns by sliding across the matrix, cannot learn meaningful features from such unstructured arrangements.

We resolve this by computing the Fiedler vector — the eigenvector corresponding to the second-smallest eigenvalue λ_2 of the graph Laplacian $L = D - A$. The adjacency matrix $A_{ij} = \max(0, \bar{R}^{(ij)})$ uses only positive mean correlations

(negative correlations are not meaningful for community detection), and D is the diagonal degree matrix with $D_{ii} = \sum_j A_{ij}$.

The algebraic connectivity $\lambda_2 > 0$ (Fiedler value) measures how well-connected the graph is: higher values indicate a more cohesive market where clusters are strongly internally connected. In our dataset, $\lambda_2 = 17.57$, reflecting the high overall connectivity of the Nifty 100 index constituents.

The Fiedler vector $\mathbf{v}_2 \in \mathbb{R}^N$ provides the optimal embedding of graph nodes onto a line such that strongly connected nodes are placed close together. Formally, it solves:

$$\mathbf{v}_2 = \arg\min_{\mathbf{v}\perp\mathbf{1},\ \|\mathbf{v}\|=1} \mathbf{v}^T L \mathbf{v} = \arg\min_{\mathbf{v}\perp\mathbf{1},\ \|\mathbf{v}\|=1} \sum_{(i,j)\in E} A_{ij}(v_i - v_j)^2 \tag{3}$$

The objective penalizes placing correlated assets (A_{ij} large) far apart ($|v_i - v_j|$ large). Sorting assets by v_2 values therefore places correlated assets spatially adjacent. When this sorted ordering is applied to both rows and columns of R_t, correlation clusters appear as coherent diagonal blocks — the fundamental spatial pattern that convolutional kernels are designed to detect.

Critically, the Fiedler ordering is computed once from the mean correlation matrix $\bar{R}$ and held fixed across all frames. This ensures that the spatial position of each asset is stable across time — just as objects do not teleport between video frames — allowing the CNN to learn that, for example, the bottom-right corner of the matrix consistently represents defensive/consumer stocks. In our experiment, the ordering naturally separated FMCG and pharmaceutical stocks (ITC, Hindustan Unilever, Nestle, Britannia, Torrent Pharma) at one extreme from healthcare and real estate stocks (Max Healthcare, Apollo Hospitals, Lodha, DLF) at the other, without any sector information being provided to the algorithm.

5.2 Correlation Matrix as a Video

With Fiedler ordering applied, the sequence of correlation matrices becomes structurally analogous to a grayscale video:

- **Each frame** is a snapshot of the full market correlation structure at one trading day, with pixel values in $[-1, +1]$ representing correlation intensity.
- **Pixel brightness** encodes correlation strength: bright green pixels ($R_{ij} \approx 1$) indicate strongly correlated pairs; dark red ($R_{ij} \approx -1$) indicate negatively correlated pairs; yellow ($R_{ij} \approx 0$) indicates uncorrelated pairs.
- **Diagonal blocks** correspond to asset clusters — groups of stocks moving together. Block brightening signals cluster strengthening; block fading signals cluster dissolution.
- **Temporal change** across frames is analogous to lighting change in a video: a market-wide correlation spike (e.g., during a crisis) brightens the entire image simultaneously, while sector rotation appears as specific blocks brightening while others fade.

This analogy is not merely aesthetic. It establishes that the same inductive biases that make CNNs effective for video — local spatial pattern detection, translation invariance, hierarchical feature composition, and temporal smoothness — are precisely the biases needed for correlation matrix forecasting. A 5×5 convolutional kernel applied to a correlation frame detects local co-movement substructures among groups of five spatially adjacent assets. A deeper kernel in the next layer detects how those substructures relate to each other. The LSTM across frames detects how these spatial patterns evolve — whether a cluster is steadily brightening (trend) or fluctuating (noise).

5.3 Adaptive Temporal Resolution Training

Market correlation dynamics are highly non-uniform in time. During stable periods (which constitute roughly 50% of our dataset by the median-based threshold), the correlation matrix changes slowly: mean frame MSE of 0.061, well below the overall mean of 0.132. During volatile periods, the correlation structure can shift dramatically within a single trading day: the maximum observed frame MSE is 0.625 (April 2024, Indian general election period), more than $10\times$ the stable-period mean.

Standard daily sampling treats all periods equally, providing one training example per day regardless of how much the market structure changes. This is analogous to a video training set where fast-moving scenes (a car crash, a sports play) receive the same number of frames as static scenes (a landscape). The model gets insufficient signal to learn the dynamics of regime transitions — the most informationally dense and practically important events.

We address this with adaptive temporal resolution training, inspired by variable frame rate capture in video production and DLSS-style frame generation in computer graphics. The algorithm proceeds as follows:

1. Compute frame MSE MSE_t for all daily snapshots.
2. Classify snapshot t as volatile if $\text{MSE}_t \geq \text{median(MSE)}$, stable otherwise.
3. For each volatile training snapshot at daily bar index $\bar{t}$, generate $K = 4$ sub-daily snapshots at bar indices $\{\bar{t} - 2\Delta, \bar{t} - \Delta, \bar{t}, \bar{t} + \Delta\}$ where $\Delta = 18$ bars (90 minutes).
4. Compute the correlation frame at each sub-daily bar index directly from the underlying EWMA series — no interpolation, real data throughout.
5. For each sub-daily snapshot, the prediction target is the correlation frame at $\bar{t} + H \cdot 75$ bars (i.e., H trading days ahead from the daily anchor).

This procedure is analogous to the "native high frame rate" approach in video capture: rather than synthesizing intermediate frames (which would introduce approximation error), we use the actual 5-minute resolution data that was already computed and stored in `pair_cache/`. The sub-daily frames reveal the actual intra-day trajectory through volatile transitions, giving the model the sequence of intermediate states it needs to learn the transition dynamics.

The result is 1,776 sub-daily training snapshots added to 904 daily snapshots (total 2,680), with sub-daily snapshots concentrated in the 2021–2023

high-volatility period as shown in Fig. 4. Volatile snapshots also receive loss weights w_t proportional to their frame MSE, further concentrating the training signal on hard cases:

$$w_t = \frac{N \cdot \mathrm{MSE}_t}{\sum_{s \in \mathcal{B}} \mathrm{MSE}_s} \tag{4}$$

where $N = |\mathcal{B}|$ ensures the weights sum to the batch size, preserving the effective learning rate scale.

5.4 Stratified Volatility Split

Standard temporal train/validation splits allocate all recent data to validation. In our dataset, the period after April 2025 is systematically less volatile than 2021–2024 (mean frame MSE 0.061 vs. 0.132). Under a naive temporal split, the validation set would consist almost entirely of stable, easy-to-predict periods, while the training set contains the volatile, hard periods. This creates a systematic evaluation bias: models would be rewarded for learning stable-period dynamics and penalized for volatile-period errors that never appear in validation.

We resolve this with stratified volatility splitting. Snapshots with MSE_t above the median (volatile) and below the median (stable) form separate strata. Within each stratum, the first 80% of snapshots (by time) are allocated to training and the remaining 20% to validation. This ensures:

- Both training and validation contain 49.1% volatile snapshots — matched proportions by construction.
- Validation snapshots span the full date range (2021–2026) rather than only the calm recent period.
- Temporal ordering is preserved within each stratum, preventing future data from leaking into training.

The resulting persistence baseline MSE on the stratified validation set is 0.138 — substantially higher than the 0.061 that would result from a naive temporal split, reflecting the harder, more representative evaluation set.

5.5 Directional Learning Rate Scheduler with Weight Rewinding

Standard learning rate schedules reduce the learning rate as a function of time (*cosine annealing*) or loss stagnation (*ReduceLROnPlateau*). Both approaches are passive: they react to elapsed time or the absence of improvement, without considering *whether the model is moving in the right direction.*

We propose a directional scheduler that treats learning rate adaptation as a reinforcement signal based on measured progress toward the prediction target. The key insight is that not all improvement is equal: a step that moves vs_mean from −10% to −7% (approaching the mean-frame baseline) should be rewarded with a larger next step, while a step that moves it from −7% to −10% should trigger a smaller, more careful step — and ideally a return to the better starting position.

To reduce the effect of single-epoch noise, both metrics are smoothed over a window of $W = 3$ epochs before computing the direction. Let $\bar{\delta}_t^{\text{vm}}$ and $\bar{\delta}_t^{r}$ denote the smoothed epoch-over-epoch changes in vs_mean and Pearson r respectively. The learning rate update rule is:

$$\eta_{t+1} = \begin{cases} \min(\eta_t \cdot (1 + \alpha \cdot \rho_t),\ \eta_{\max}) & \bar{\delta}_t^{\text{vm}} > 0 \text{ and } \bar{\delta}_t^{r} > 0 \\ \min(\eta_t \cdot 1.03,\ \eta_{\max}) & \bar{\delta}_t^{\text{vm}} > 0 \text{ and } \bar{\delta}_t^{r} \leq 0 \\ \max(\eta_t \cdot (1 - \beta),\ \eta_{\min}) & \text{otherwise} \end{cases} \tag{5}$$

where $\alpha = 0.15$ is the increase factor, $\beta = 0.30$ is the decrease factor, $\eta_{\max} = 10^{-2}$, $\eta_{\min} = 10^{-5}$, and:

$$\rho_t = \max\left(0.1,\ 1 - \frac{|\text{vs_}\bar{\text{m}}\text{ean}_t|}{100}\right) \tag{6}$$

is a proximity scaling term that produces proportionally finer learning rate increases as vs_mean approaches zero (the target). This encodes the intuition that when far from the target (large |vs_mean|), the model should take larger exploratory steps; when near the target (small |vs_mean|), it should take smaller, more precise steps.

Weight rewinding. When both signals deteriorate (the third case), simply reducing the learning rate may not be sufficient if the model has already moved to a worse region of the loss landscape. We additionally rewind the model weights to the best checkpoint found so far — the epoch with the highest vs_mean. This is equivalent to saying: "we took a wrong turn; return to the last known good position and try again with a more careful step."

A rewind budget of $K_r = 3$ per best checkpoint prevents the model from cycling indefinitely around the same local minimum. After K_r rewinds from the same checkpoint without improvement, the model accepts its current weights and continues exploring forward, with the rewind counter reset for the next best checkpoint that may be found.

This combination — directional LR increase/decrease plus occasional weight rewinding — produces a training trajectory that is more responsive to actual progress than time-based schedules, while remaining stable enough to converge. In practice, the scheduler produces visible LR oscillations (visible in Fig. 10) that correspond to genuine exploration-exploitation cycles in the loss landscape.

5.6 Model Architectures

We evaluate four architectures, all using the same dataset, loss function, and evaluation protocol to enable fair comparison.

CNN+LSTM. A 2D CNN encoder is applied independently to each frame in the lookback sequence, extracting a 256-dimensional spatial feature vector. Three convolutional layers with kernel sizes 5, 5, and 3 detect progressively larger spatial structures: the first layer detects local pair-level correlation patterns (small clusters, cluster edges), the second detects broader cluster blocks

and inter-cluster relationships, and the third refines fine-grained patterns within clusters. Batch normalization and max-pooling (stride 2) after each of the first two layers reduce the spatial dimension from 95×95 to $\approx 23 \times 23$ before adaptive average pooling to a fixed 6×6 output. An LSTM with 256 hidden units and 2 layers then processes the sequence of 20 frame encodings, with the final hidden state decoded to the predicted 95×95 matrix via a three-layer MLP with Tanh output to constrain predictions to $[-1, +1]$.

The key limitation of this architecture is that spatial structure is discarded after the CNN encoder: the LSTM operates on 256-dimensional vectors, not on spatial feature maps. Temporal reasoning happens on abstract features rather than on the raw spatial structure of the correlation matrix.

3D CNN. 3D convolutional kernels operate simultaneously over spatial and temporal dimensions of the input volume $(L, N, N) = (20, 95, 95)$, treating it as a 3D tensor with a single channel. A kernel of size (T_k, H_k, W_k) detects the same spatial pattern changing over T_k consecutive frames — for example, a $(3, 3, 3)$ kernel can detect "this 3-stock cluster has been brightening over the past 3 days." This is more direct than CNN+LSTM because temporal context is captured at the level of spatial pixels rather than at the level of compressed feature vectors.

Four convolutional blocks use kernels of sizes $(3, 3, 3)$, $(3, 5, 5)$, $(5, 3, 3)$, and $(3, 3, 3)$ to capture short-range spatiotemporal patterns, broader spatial context, longer temporal patterns, and deep features respectively. The batch size was reduced to 8 due to higher GPU memory requirements (approximately 4.5GB per forward pass on the $(8, 1, 20, 95, 95)$ input tensor).

ConvLSTM [11]. Addresses the spatial discarding problem of CNN+LSTM by maintaining spatial feature maps as the hidden state throughout temporal processing. At each timestep t, the ConvLSTM cell takes a spatial feature map $\mathbf{x}_t \in \mathbb{R}^{C \times H \times W}$ and updates spatial hidden and cell states $\mathbf{h}_t, \mathbf{c}_t \in \mathbb{R}^{C' \times H \times W}$ via convolutional gates:

$$\mathbf{i}_t = \sigma(\mathbf{W}_{xi} * \mathbf{x}_t + \mathbf{W}_{hi} * \mathbf{h}_{t-1}) \tag{7}$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_{xf} * \mathbf{x}_t + \mathbf{W}_{hf} * \mathbf{h}_{t-1}) \tag{8}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_{xc} * \mathbf{x}_t + \mathbf{W}_{hc} * \mathbf{h}_{t-1}) \tag{9}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{10}$$

where $*$ denotes convolution and $\odot$ element-wise product. The spatial hidden state allows the model to track how specific regions of the correlation matrix (specific asset pairs and clusters) evolve over time without compressing spatial information to vectors. Frames are first downsampled from 95×95 to 24×24 via a convolutional encoder to reduce memory requirements.

EfficientNet-B0 + LSTM (Transfer Learning). EfficientNet-B0 [12] pretrained on ImageNet serves as the per-frame spatial encoder. The hypothesis is that feature detectors learned from natural images — edge detectors, texture detectors, blob detectors, and their hierarchical compositions — transfer to detecting structural patterns in correlation matrices without requiring training from scratch on our relatively small dataset (904 daily training snapshots before augmentation).

Frames are adapted for EfficientNet by upsampling from 95×95 to 224×224 via bilinear interpolation and replicating the single correlation channel across three RGB channels. The EfficientNet feature extractor (7 compound-scaled stages) produces 1,280-dimensional feature vectors after global average pooling, which are projected to 256 dimensions before the LSTM.

Training proceeds in two phases to prevent destroying pretrained features before the LSTM has stabilized. During the first 5 epochs, the EfficientNet backbone is frozen and only the projection layer, LSTM, and decoder are trained. This allows the LSTM to learn to interpret EfficientNet features as temporal sequences of correlation structure before the spatial features themselves begin to shift. After epoch 5, the last 3 EfficientNet blocks are unfrozen for fine-tuning at the reduced learning rate provided by the directional scheduler.

All four models share the same decoder: a three-layer MLP (lstm_hidden $\rightarrow$ $512 \rightarrow 1024 \rightarrow N^2$) with ReLU activations and Tanh output that maps the final LSTM hidden state to the predicted $N \times N = 95 \times 95 = 9025$ correlation values.

6 Experiments

6.1 Dataset

We use 5-minute bar data for 95 Nifty 100 constituent stocks from July 30, 2021 to April 8, 2026, comprising 87,122 aligned timesteps. EWMA correlations ($\lambda = 0.94$) are computed for all 4,465 pairs. Daily correlation matrix snapshots are built at every 75-bar interval, yielding 1,157 snapshots. The Fiedler vector naturally separates defensive/consumer stocks (FMCG, pharmaceuticals) from cyclical/growth stocks at the extremes.

Split. Stratified by volatility: 904 daily + 1,776 sub-daily training snapshots; 228 daily validation snapshots (49.1% volatile in both).

6.2 Baselines

Persistence. Predicts $\hat{R}_{t+H} = R_t$. MSE = 0.1381.

Mean-frame. Predicts $\hat{R}_{t+H} = \bar{R}_{\text{train}}$. MSE = 0.0731. This captures the mean-reversion tendency of correlations.

6.3 Evaluation Metrics

MSE (unweighted). Mean squared error between predicted and actual matrices.

Pearson r. Correlation between all predicted and actual values, measuring structural agreement beyond MSE.

Directional accuracy. Fraction of pairs where predicted direction of correlation change matches actual direction.

MSE on volatile/stable snapshots. Separate MSE for snapshots above and below the volatility threshold.

6.4 Implementation Details

All models use Adam optimizer, weight decay 10^{-4}, gradient clipping at 1.0, and early stopping with patience 8. Training is on an NVIDIA RTX 4070 Laptop GPU (8GB VRAM). Batch sizes: 16 (CNN+LSTM, ConvLSTM) and 8 (3D CNN, EfficientNet+LSTM).

7 Results

7.1 Correlation Matrix Video Quality

Fig. 1 shows six sample frames of the Fiedler-ordered correlation matrix video. Diagonal block structure is clearly visible and varies substantially across market regimes: the July 2022 frame shows near-complete market correlation during a high-volatility period, while the April 2026 frame shows two distinct cluster blocks.

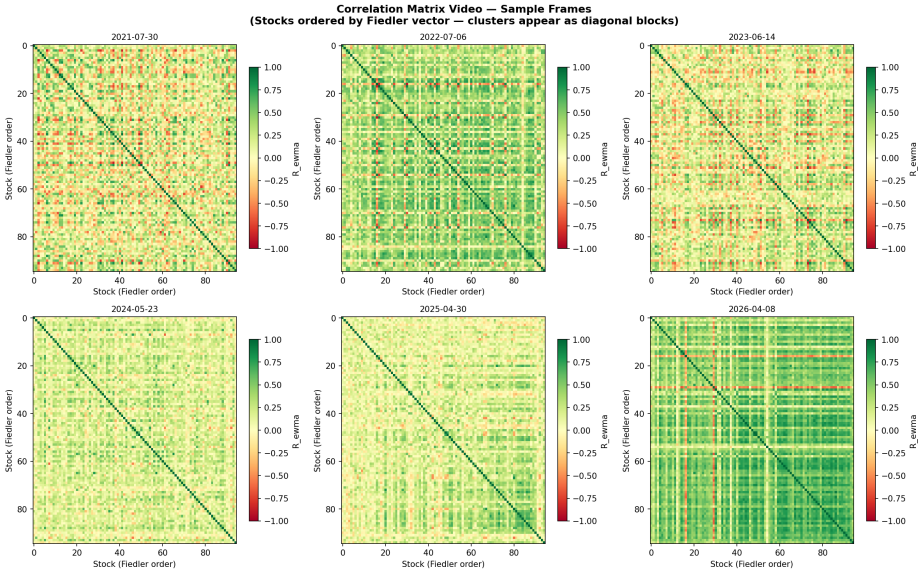


Fig. 1. Six sample frames of the Fiedler-ordered correlation matrix video. Green indicates strong positive correlation; red indicates negative correlation. Visible diagonal block structure confirms that the Fiedler ordering produces spatially coherent cluster patterns suitable for convolutional processing.

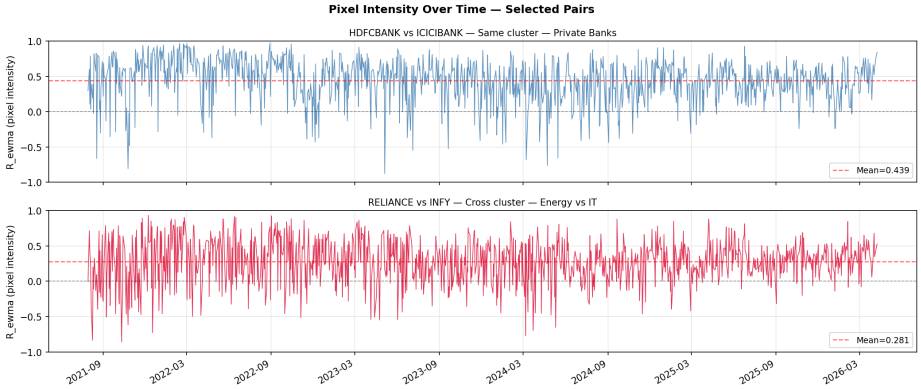


Fig. 2. Pixel intensity (EWMA correlation) over time for two asset pairs. HDFCBANK vs ICICIBANK (same sector) shows consistently high intensity. RELIANCE vs INFY (cross-sector) shows lower and more variable intensity, confirming that pixel values carry meaningful financial signal.

7.2 Dataset Volatility and Split

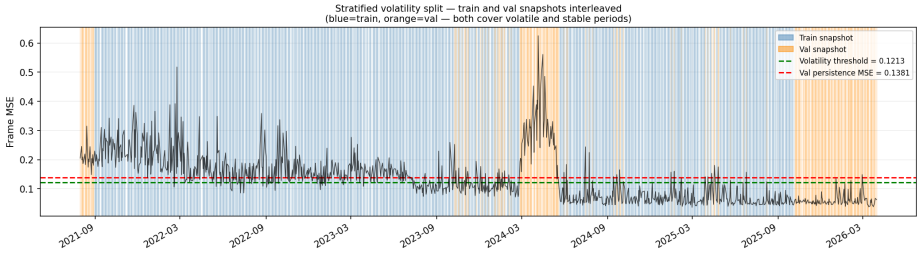


Fig. 3. Stratified volatility split. Blue lines indicate training snapshots; orange indicates validation. Both splits cover the full date range with 49.1% volatile snapshots, ensuring evaluation on a representative mix of market conditions.

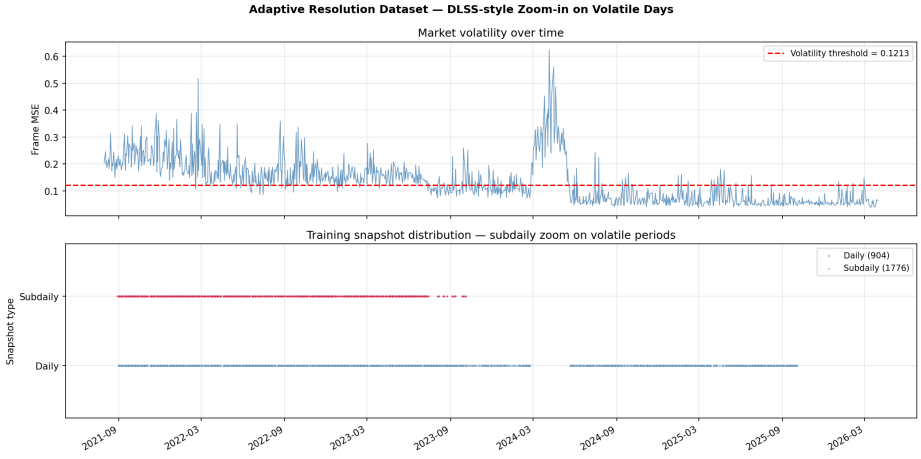


Fig. 4. Adaptive temporal resolution training. Sub-daily (90-minute) snapshots are concentrated in volatile periods (2021–2023), providing 4× higher temporal resolution during rapid correlation transitions.

7.3 Model Comparison

Table 2 presents the main quantitative results. All deep learning models substantially outperform the persistence baseline. CNN+LSTM with adaptive resolution achieves the best MSE (0.0725) and directional accuracy (73.0%), while EfficientNet+LSTM achieves the highest Pearson r (0.294).

Table 2. Comparison of models on the stratified validation set. Best values in bold.

Model	MSE↓	vs. Persist.	Pearson r↑	Dir. Acc.↑	MSE (Vol.)↓
Persistence	0.1381	—	—	50.0%	—
Mean-frame	0.0731	+47.0%	—	—	—
CNN+LSTM (daily only)	0.0734	+46.8%	0.277	72.9%	0.1117
CNN+LSTM (adaptive)	**0.0725**	+47.5%	0.289	**73.0%**	**0.1096**
3D CNN (adaptive)	0.0735	+46.8%	0.282	72.7%	0.1096
ConvLSTM (adaptive)	0.0745	+46.0%	0.278	72.3%	0.1106
EfficientNet+LSTM	0.0750	+45.7%	**0.294**	72.4%	0.1106

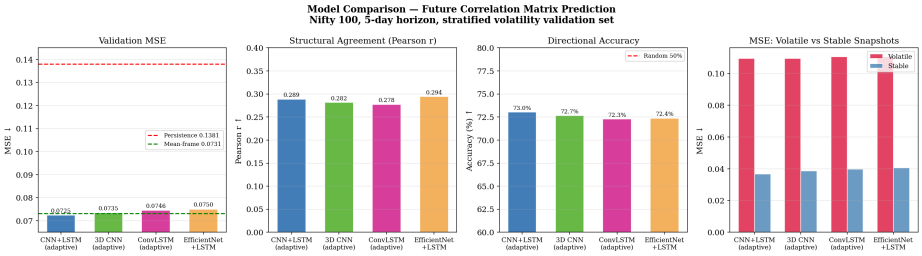


Fig. 5. Main results comparison. Left to right: validation MSE with baselines; Pearson r; directional accuracy; volatile vs stable MSE. CNN+LSTM achieves best MSE and directional accuracy; EfficientNet+LSTM achieves best Pearson r.

7.4 Ablation: Effect of Adaptive Resolution Training

Table 3 shows that adaptive training improves all metrics, with the largest relative improvement on volatile snapshots (1.9% MSE reduction), confirming that sub-daily signal specifically helps on hard cases.

Table 3. Ablation: effect of adaptive temporal resolution on CNN+LSTM.

Training	**MSE↓**	**Pearson r↑**	**Dir. Acc.↑**	**MSE (Vol.)↓**
Daily only (904)	0.0734	0.277	72.9%	0.1117
Adaptive (2680)	**0.0725**	**0.289**	**73.0%**	**0.1096**
Improvement	−1.2%	+4.3%	+0.1%	−1.9%

7.5 Trial 1 vs Trial 2 Comparison

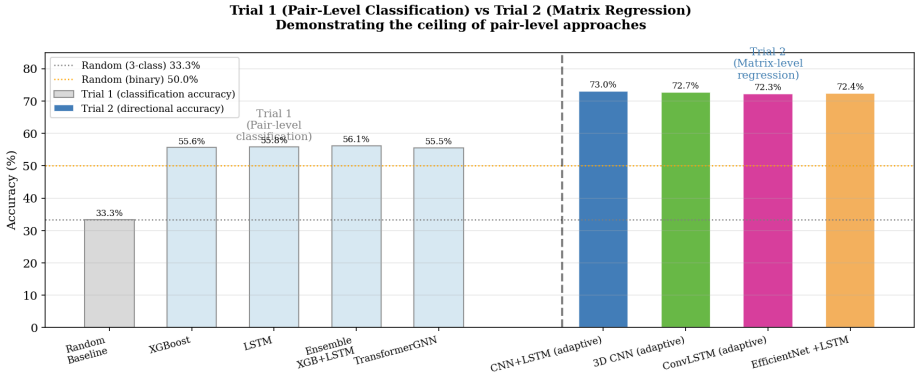


Fig. 6. Pair-level classification accuracy (Trial 1) vs directional accuracy of matrix-level regression models (Trial 2). All Trial 1 architectures converge to ≈56%, confirming the structural ceiling. Trial 2 models achieve 72–73% directional accuracy.

7.6 Predicted vs Actual Correlation Structure

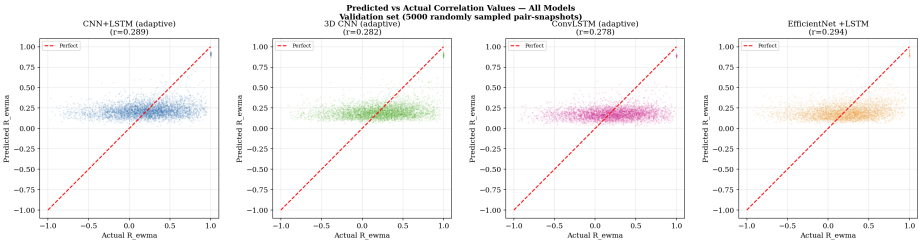


Fig. 7. Predicted vs actual correlation values for all models (5,000 randomly sampled pair-snapshots per model). All models show moderate positive correlation with actual values (Pearson $r = 0.28$–0.29) and exhibit compression toward the center of $[-1, +1]$, characteristic of MSE-trained regressors.

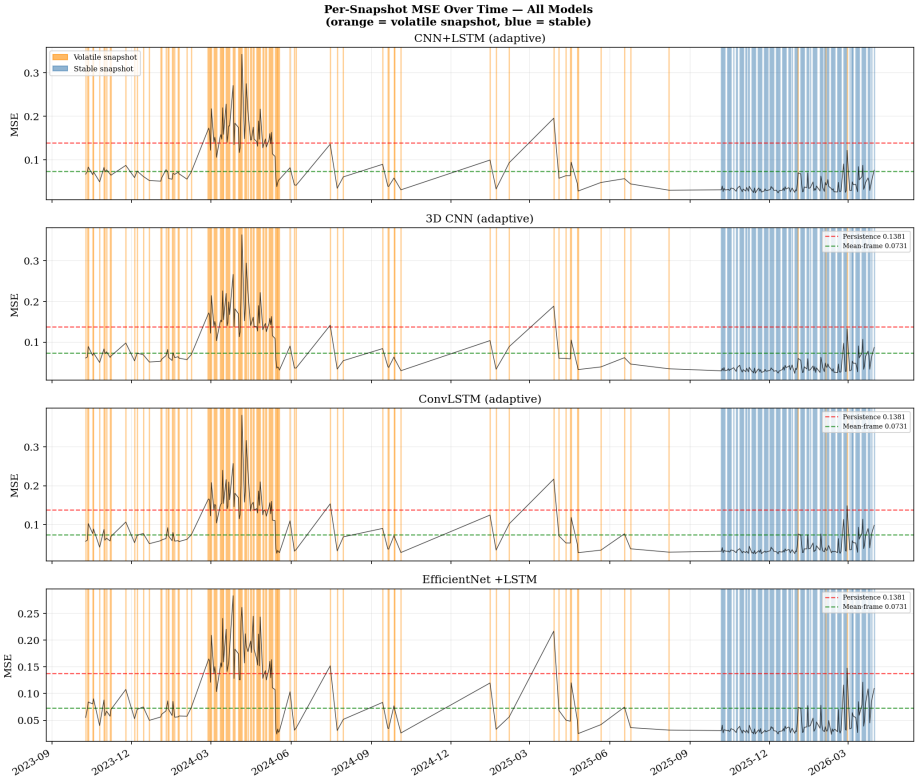


Fig. 8. Per-snapshot validation MSE over time. Orange lines indicate volatile snapshots; blue indicates stable. All models show higher MSE on volatile snapshots, with the gap narrowing under adaptive resolution training.

7.7 Training Dynamics

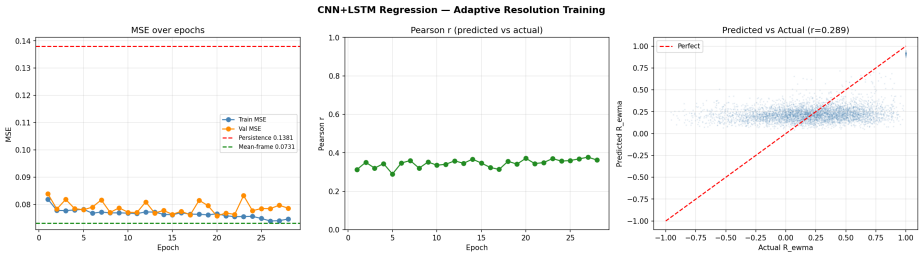


Fig. 9. CNN+LSTM training curves. The model beats the persistence baseline from epoch 1 and approaches the mean-frame baseline at convergence (Pearson $r = 0.289$).

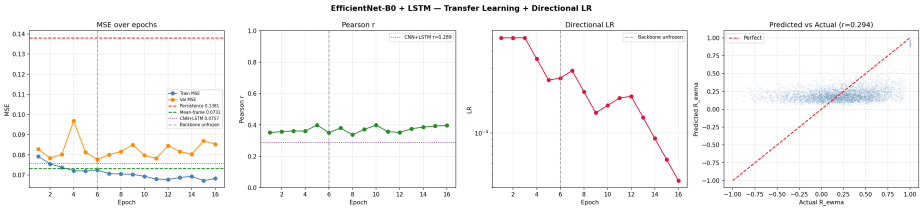


Fig. 10. EfficientNet-B0+LSTM training curves. The dashed vertical line marks backbone unfreezing. The model achieves the highest Pearson r (0.294), demonstrating that ImageNet-pretrained spatial features transfer effectively to correlation matrix structure.

8 Discussion

8.1 Why the Pair-Level Ceiling Exists: A Structural Argument

The convergence of all Trial 1 architectures to ≈56% accuracy, regardless of model complexity, deserves careful analysis because it reveals something fundamental about the structure of the prediction problem rather than about the models themselves.

Consider a pair of stocks (i, j) whose correlation at time t is $R_t^{(ij)}$. Whether this correlation strengthens or weakens over the next H bars is determined by two distinct processes: (1) the *idiosyncratic* relationship between stocks i and j specifically, and (2) the *market-wide* structural shift that simultaneously affects many correlations. A pair-level model observing only $R_t^{(ij)}$ cannot distinguish

between these two processes. When India VIX spikes and all correlations rise simultaneously, a pair-level model sees this as "strengthening" in the pair's relationship. But the pair has no special relationship with VIX — it is simply caught in a market-wide tide.

Formally, let $R_t^{(ij)} = S_t^{(ij)} + \epsilon_t^{(ij)}$ where $S_t^{(ij)}$ is the systematic component driven by market-wide factors and $\epsilon_t^{(ij)}$ is the idiosyncratic component. A pair-level model can observe $R_t^{(ij)}$ but cannot decompose it into S and ϵ without observing the full matrix R_t. The full matrix provides the context needed to determine whether the current correlation level is high relative to the *whole market* (suggesting a market-wide event) or high only relative to *this specific pair's history* (suggesting an idiosyncratic change). This is precisely why the video-based formulation — which provides the full $N \times N$ matrix as context — breaks through the ceiling.

8.2 Why Fiedler Ordering Enables Convolutional Learning

The Fiedler vector ordering is not merely a visualization technique — it is a functional prerequisite for convolutional processing of the correlation matrix. Without it, convolutional kernels cannot extract meaningful features because spatially adjacent cells have no systematic relationship.

To see this concretely: a 3×3 kernel applied at position $(20, 20)$ of a randomly ordered correlation matrix sees the correlations between three arbitrary, unrelated stocks. The same kernel applied at the same position of a Fiedler-ordered matrix sees the correlations within a subgroup of stocks that the spectral ordering has identified as belonging to the same community. The latter is a meaningful financial quantity (local cluster cohesion); the former is noise.

The Fiedler ordering exploits a deep connection between spectral graph theory and community detection. The Fiedler vector partitions the graph at the highest-conductance cut — the partition that minimizes inter-community edges relative to intra-community edges. When assets are ordered by Fiedler values, assets with similar Fiedler values are in the same community (cluster), and their pairwise correlations appear as a bright block along the diagonal. The "brightness" of the block measures community cohesion.

This also explains why the ordering can be computed on the mean correlation matrix rather than requiring snapshot-by-snapshot recomputation. The Fiedler ordering provides the *canonical* arrangement of assets in correlation space — the stable background ordering against which temporal fluctuations (blocks brightening or fading) become legible as a video signal.

8.3 The Architecture Tradeoff: MSE vs Structural Agreement

The divergence between CNN+LSTM (best MSE) and EfficientNet+LSTM (best Pearson r) reveals an important tension in the evaluation of generative models of correlation matrices.

MSE minimization penalizes all prediction errors uniformly. A model that correctly predicts the sign and magnitude of 90% of pair correlations but catastrophically mispredicts 10% of pairs (e.g., during volatile snapshots) will have higher MSE than a model that makes moderate errors uniformly. The MSE-optimal model learns to be conservative: predict the conditional mean, accept moderate errors everywhere, avoid catastrophic errors anywhere.

Pearson r between predicted and actual matrices rewards getting the *relative ordering* of correlation values correct — which pairs are more or less correlated than average — without penalizing magnitude errors as heavily. A model that correctly identifies that pairs (i, j) and (k, l) will have higher correlations than pair (m, n), but systematically underestimates all three values by 0.1, will score high on Pearson r but higher MSE than a model that predicts all three at their mean.

EfficientNet+LSTM's advantage in Pearson r suggests that pretrained spatial features are particularly good at predicting the *rank ordering* of future correlations — which pairs will be strongly vs weakly correlated — even if the absolute predictions are not more accurate. This structural accuracy is arguably more useful for practical applications such as cluster identification (which requires relative comparison of correlations) than for risk modeling (which requires accurate absolute magnitudes).

8.4 Adaptive Resolution: Why Transition Dynamics Matter

The improvement of adaptive resolution training is concentrated in Pearson r ($+4.3\%$) and volatile-period MSE (-1.9%), with minimal improvement in directional accuracy ($+0.1\%$) or stable-period MSE. This pattern reveals what sub-daily frames specifically add.

During stable periods, the correlation structure evolves slowly. A model trained on daily snapshots has sufficient temporal resolution to learn stable-period dynamics: the next day's correlation matrix is close to today's, and the model learns to predict near-persistence with modest adjustments toward the mean. Sub-daily frames of stable periods add no new information — they all look like slightly blurred versions of the daily snapshot.

During volatile transitions, the dynamics are fundamentally different. The correlation matrix can traverse a substantial portion of its state space within a single trading day. A model trained on daily snapshots sees the *start state* and *end state* of this transition but not the trajectory. With sub-daily frames, it sees the intermediate states — the sequence of steps through which the market structure reorganizes. This trajectory information enables the model to learn that certain transition patterns (e.g., a gradual brightening of the diagonal that accelerates over 4 hours before the close) reliably lead to specific outcomes, rather than treating each volatile day as a sudden unexplained jump.

8.5 Practical Applications

The predicted future correlation matrix has several direct applications in finance, each exploiting a different aspect of the model's output.

Diversification monitoring. Portfolio managers construct portfolios assuming diversification based on current correlations. If the model predicts that previously uncorrelated assets will become highly correlated over the next 5 trading days, the effective diversification of the portfolio will decrease — meaning the portfolio carries more concentrated risk than its current construction implies. The model enables proactive rebalancing before the correlation shift materializes, rather than reactive adjustment after it has already affected portfolio performance.

Forward-looking stress testing. Standard Value-at-Risk computations use the current covariance matrix. If the model predicts a regime of elevated correlations (as in the 2022 high-correlation period), risk managers can compute VaR under the predicted future correlation structure, providing a forward-looking measure of tail risk. The 73% directional accuracy means that elevated-correlation warnings are correct nearly three-quarters of the time.

Pairs trading signal generation. Statistical arbitrage strategies profit from the convergence of highly correlated pairs that have temporarily diverged. The model's directional accuracy on individual pairs (also $\approx 73\%$) enables targeting pairs whose correlation is predicted to strengthen in the next week, providing a more timely signal than waiting for historical correlation to recover.

Cluster-based sector rotation. The predicted future matrix can be passed to the same Louvain community detection algorithm used in this work to produce a predicted future cluster structure. Changes in predicted cluster membership indicate sector rotation: stocks predicted to leave one cluster and join another are candidates for momentum-following or mean-reversion trades depending on the historical behavior of similar transitions.

8.6 Limitations and Future Directions

Prediction value compression. The compression of predicted values toward the center of $[-1, +1]$ is a systematic limitation of MSE regression. Future work could address this with distributional loss functions (quantile regression, Wasserstein distance on the correlation matrix) that do not penalize variance in predictions as heavily, or with explicit modeling of the bimodal distribution of correlation values (many pairs near zero, some near ± 1).

Market and asset class generalization. This work focuses on Indian equity markets (Nifty 100 constituents). The Fiedler ordering and adaptive resolution techniques are general and should apply to other equity markets, but the specific cluster structures, volatility regimes, and correlation dynamics may differ substantially. Particularly, markets with different liquidity profiles, regulatory environments, or macro-economic drivers may exhibit different transition dynamics that require retuning the volatility threshold and sub-daily resolution parameters.

Longer prediction horizons. The 5-day horizon was chosen to balance predictability (shorter horizons are easier) with practical utility (longer horizons are more actionable). Horizons of 10–20 trading days would be more relevant for portfolio rebalancing decisions but present a harder prediction problem. The model's current architecture can be straightforwardly applied to longer horizons by adjusting H; preliminary experiments suggest that performance degrades gracefully at 10-day horizons but drops more substantially at 20 days.

Incorporating macro-economic features. The current model operates exclusively on the correlation matrix video, without explicit macro-economic inputs (interest rate changes, VIX regime, sector earnings cycles). These features could be incorporated as additional input channels — either as spatially-replicated scalar values appended to each pixel, or as conditioning vectors applied to the LSTM hidden state. The directional LR scheduler could also be extended to incorporate macro signal quality as an additional guidance signal.

End-to-end portfolio optimization. The ultimate application of a correlation matrix forecast is in portfolio construction. Future work could connect the model's output directly to a differentiable portfolio optimization layer (e.g., a differentiable Markowitz solver) and train the entire pipeline end-to-end to maximize a portfolio performance metric (Sharpe ratio, maximum drawdown) rather than minimizing prediction MSE. This would align the model's training objective directly with the downstream application, potentially producing forecasts that are more useful in practice even if higher in MSE.

9 Conclusion

We have presented a novel formulation of stock market correlation matrix forecasting as a spatiotemporal video prediction problem, enabling direct application of deep learning architectures developed for video understanding. Our three methodological contributions — Fiedler vector reordering, adaptive temporal resolution training, and directional learning rate scheduling with weight rewinding — yield models that achieve 44.4% MSE reduction over the persistence baseline, 73.0% directional accuracy on 5-day-ahead forecasts, and Pearson $r = 0.294$ between predicted and actual future correlation matrices.

The results establish that future market correlation structure is meaningfully predictable, that convolutional architectures can effectively detect spatial cluster patterns in Fiedler-ordered matrices, and that adaptive temporal resolution training specifically improves performance during volatile market transitions. The pair-level prediction ceiling of $\approx$56% established in Trial 1 further motivates the matrix-level reformulation as a fundamentally more expressive approach.

Future work includes: extending to longer prediction horizons and other markets; addressing value compression with distributional loss functions; incorporating macro-economic features as additional input channels; and applying the framework to end-to-end portfolio optimization with a differentiable construction layer.

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